BUDAPEST SEMESTERS IN MATHEMATICS
ADVANCED DIRECTED READING PROGRAM

Morse Theory

Monica Sing

Supervisor: Prof. András Stipsicz, Balázs Attila Forman

BSM-ADRP-2026-xx

Autumn 2026

Webpage: https://drp.elte.hu/

Morse Theory

Monica Sing*

Abstract

This paper present a self-contained introduction to Morse theory, emphasizing the relationship between critical points of smooth functions and the topology of manifolds. Beginning with the differential–topological background, we develop the Morse lemma and analyze the topology of interval sublevel sets both in the absence of critical points and containing non-degenerate critical points. These local results lead to a global description of manifolds via CW-complex and handlebody decompositions. We then derive the Morse inequalities using relative homology and Euler characteristic arguments and conclude with a brief discussion of further directions with Morse homology.

Contents

1 Preliminary background for the Morse Lemma **2**
- 1.1 Differential Structure 2
- 1.2 Critical Point Theory 3
- 1.3 Vector Fields 6

2 Morse Functions and the Morse Lemma **9**

3 Interval sublevel set containing no critical points **14**

4 Interval sublevel set containing a non-degenerate critical point **16**

5 CW-complex and Handlebody decomposition **21**
- 5.1 Handlebody Decomposition for 3-Manifolds 26
- 5.2 Handlebody Decomposition for 4-Manifolds 26

6 Morse Inequalities **29**

7 Further Explorations: Morse Homology **35**

8 Acknowledgments **37**

9 Citations **37**

*Department of Mathematics, Princeton University. Email: `monicasing@princeton.edu`.

1 Preliminary background for the Morse Lemma

This section reviews the differential–topological background required for the Morse Lemma. We recall basic properties of smooth manifolds and tangent bundles, introduce the theory of critical points of smooth functions, and review vector fields and their flows, which will later be used to give geometric interpretations of Morse-theoretic results.

1.1 Differential Structure

Definition 1.1 (Tangent Vector)
Let M be a differentiable manifold, let $p \in M$, and let $\gamma_1, \gamma_2 : (-\epsilon, \epsilon) \to M$ be curves. $\gamma_1 \sim \gamma_2$ iff $\gamma_1(0) = \gamma_2(0) = p$ and for any local coordinate chart $\phi : M \to \mathbb{R}^n$,

$$(\phi \circ \gamma_1)'(0) = (\phi \circ \gamma_2)'(0),$$

that is, their velocities at $t = 0$ agree. A tangent vector $v \in T_pM$ is the the equivalence class $[\gamma]$ of all such curves that have the same velocity at p.

Alternatively, we can define tangent vectors in the setting of differentiation, ...

Definition 1.2 (Tangent Space)
The tangent space of M at p, T_pM is the set of all tangent vectors at p.

Definition 1.3 (Tangent Bundle)
The tangent bundle of M is the disjoint union of all tangent spaces of M,

$$TM = \bigsqcup_{p \in M} T_pM = \bigcup_{p \in M} \{(p, y) \mid y \in T_pM\},$$

that is equipped with the natural topology as follows.

Remark 1.4 (Topology of the Tangent Bundle)
The topology of TM is given by the continuous natural projection

$$\begin{aligned} \pi : TM &\to M \\ (p, y) &\to p. \end{aligned}$$

For each open set $U \subseteq M$ and its chart, $(U, \psi : U \to \mathbb{R})$, the induced chart on TM is

$$\begin{aligned} \tilde{\phi} : \pi^{-1}(U) &\to \mathbb{R}^{2n} \\ (p, y) &\mapsto (\phi(p), d\phi(y)). \end{aligned}$$

The topology on TM is the unique topology making all such $\tilde{\phi}$ homeomorphisms onto open subsets of $\mathbb{R}^{2n}$. The resulting collection $\{(\pi^{-1}(U), \tilde{\varphi})\}$ forms a smooth atlas on TM, making it a $2n$-dimensional smooth manifold.

Definition 1.5 (Tangent Map)
Given a smooth map $f : M \to N$, a curve $\gamma(t) : \mathbb{R} \to M$ with $\gamma(0) = p, \gamma'(0) = v \in T_pM$, the induced map $f_* : T_pM \to TN_{f(p)}$ is a linear transformation that acts on v by

$$f_*(v) = (f \circ \gamma)'(0).$$

The tangent map is also known as the differential, the Jacobian, or the pushforward.

Definition 1.6 (Exterior Derivative)
Given a smooth function $f : M \to \mathbb{R}$, the exterior derivative df is a cotangent vector field on M assigning to each point $p \in M$ a cotangent vector acting on T_pM, in particular, given some vector $v \in T_pM$, $df(v)$ is the directional derivative of f in the direction of v.

More generally, for any vector field X, $df(X) = X(f)$, where $X(f)$ is the the directional derivative of f along X at each point.

Definition 1.7 (Jacobian)
Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a differentiable function, and define f_i as

$$\mathbf{f}(x_1, x_2, \ldots, x_n) = \begin{bmatrix} f_1(x_1, \ldots, x_n) \\ f_2(x_1, \ldots, x_n) \\ \vdots \\ f_m(x_1, \ldots, x_n) \end{bmatrix}.$$

Then the Jacobian matrix J is an $m \times n$ matrix s.t. $(J)_{ik} = \frac{\partial f_i}{\partial x_k}$, that is,

$$\mathbf{J} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}.$$

1.2 Critical Point Theory

Definition 1.8 (Critical Point)
Given a smooth real-valued function $f : M \to \mathbb{R}$ on a manifold M, a point $p \in M$ is a critical point if the tangent map $f_* : T_pM \to T\mathbb{R}_{f(p)}$ is zero. That is, if we choose a local coordinate system $(x^1, \ldots, x^n)$ in a neighborhood U of p, p is a critical point when

$$\frac{\partial f}{\partial x^1} = \cdots = \frac{\partial f}{\partial x^n} = 0$$

Definition 1.9 (Non-Degenerate Critical Point)
A critical point p is non-degenerate if and only if the matrix

$$A = \left(\frac{\partial^2 f}{\partial x^i \partial x^j}(p) \right)$$

is non-singular, that is, $\det(A) \neq 0$.

Definition 1.10 ((Symmetric) Bilinear Form)
Let V be a Vector Space of dimension n over a field $\mathbb{F}$. A map $B : V \times V \to \mathbb{F}$ is a (symmetric) bilinear form on V if $\forall u, v, w \in V, \forall \lambda \in \mathbb{F}$:

(1) $B(u + v, w) = B(u, w) + B(v, w)$
(2) $B(\lambda u, v) = \lambda B(u, v)$
(3) $B(u, v) = B(v, u)$

Definition 1.11 (Hessian)
The Hessian can be defined as a symmetric bilinear function f_{**} on T_pM, such that, for $v, w \in T_pM$,

$$f_{**}(v,w) = \tilde{v}_p(\tilde{w}(f))$$

where $\tilde{v}, \tilde{w}$ are vector field extensions of v and w, and $\tilde{v}_p = v$. This essentially represents the 2nd order directional derivatives.

We have to extend v, w into vector fields as we can not simply differentiate a single vector across a manifold. In order to compute $\tilde{v}_p(\tilde{w}(f))$, in which we apply one directional derivative (along $\tilde{v}_p$) to the result of another ($\tilde{w}(f)$), $\tilde{w}$ must be defined on a neighborhood of p.

Specifically, we can extend tangent vectors to the constant vector field. Let $(x^1, \ldots, x^n)$ be the local coordinate system in a neighborhood U of p, and let

$$v = \sum_i a_i \frac{\partial}{\partial x^i}\bigg|_p, \quad w = \sum_j b_j \frac{\partial}{\partial x^j}\bigg|_p,$$

be two tangent vectors at p, where a_i's, b_j's are constants. Then, define the extensions of v and w to vector fields on U as follows,

$$\tilde{v} = \sum_i a_i \frac{\partial}{\partial x^i}, \quad \tilde{w} = \sum_j b_j \frac{\partial}{\partial x^j}.$$

Therefore,

$$\begin{aligned}
f_{**}(v,w) &= \tilde{v}_p(\tilde{w}(f)) \\
&= \tilde{v}(\tilde{w}(f))(p) \\
&= \tilde{v}(\sum_j b_j \frac{\partial f}{\partial x^j})(p) \\
&= \sum_{i,j} a_i b_j \frac{\partial^2 f}{\partial x^i \partial x^j}(p)
\end{aligned}$$

and we have that $H_{ij} = \frac{\partial^2 f}{\partial x^i \partial x^j}$ is a bilinear function w.r.t. the basis $\frac{\partial}{\partial x^1}\Big|_p, \ldots, \frac{\partial}{\partial x^n}\Big|_p$.

Finally,

$$\begin{aligned}
f_{**}(v,w) - f_{**}(w,v) &= \tilde{v}_p(\tilde{w}(f)) = \tilde{w}_p(\tilde{v}(f)) \\
&= [\tilde{v}, \tilde{w}]_p(f) \\
&= df_p([\tilde{v}, \tilde{w}]) \\
&= 0
\end{aligned}$$

Therefore, $f_{**}(v,w)$ is symmetric.

Remark 1.12
The final step is a consequence of $df_p = 0$ by p being a critical point of f, where df_p is the exterior derivative of f at p. The definition of the exterior derivative is given in Def 1.6.

Remark 1.13
The Lie Bracket $[\tilde{v}, \tilde{w}] = \tilde{v}(\tilde{w}(f)) - \tilde{w}(\tilde{v}(f))$ *is itself a vector field on* TM. *Evaluating it at* p *gives us some tangent vector at* p, $[\tilde{v}, \tilde{w}]_p \in T_pM$

Lemma 1.14
Non-degeneracy of critical points is independent from the choice of coordinates.

Proof. Let p be a non-degenerate critical point of a map $f : M \to \mathbb{R}$. Our goal is to show that given any choice of local coordinates $\vec{x} = (x^1, \dots, x^n)$, $\vec{u} = (u^1, \dots, u^n)$ on neighborhoods of p, the determinant of the Hessian obtained from $\vec{x}, \vec{u}$ satisfies the following:

$$\det(H_{\vec{x}}) = 0 \iff \det(H_{\vec{u}}) = 0$$

.

Under $\vec{x}$, the Hessian is given by,

$$H_{ij} = \frac{\partial^2 f}{\partial x^i \partial x^j}.$$

Applying a change of coordinates $x^i \mapsto u^i(\vec{x})$, we can apply the chain rule to obtain a transformation law,

$$\begin{aligned}
\frac{\partial}{\partial u^j}\left(f(x(u)\right) &= \sum_l \left(\frac{\partial f(x(u))}{\partial x^l}\frac{\partial x^l}{\partial u^j}\right) \\
\frac{\partial}{\partial u^i}\left(\frac{\partial}{\partial u^j}\left(f(x(u))\right)\right) &= \frac{\partial}{\partial u^i}\sum_l \left(\frac{\partial f(x(u))}{\partial x^l}\frac{\partial x^l}{\partial u^j}\right) \\
&= \sum_{k,l}\left(\frac{\partial^2 f(x(u))}{\partial x^k \partial x^l}\frac{\partial x^k}{\partial u^i}\frac{\partial x^l}{\partial u^j}\right) + \sum_l \left(\frac{\partial f(x(u))}{\partial x^l}\frac{\partial^2 x^l}{\partial u^i \partial u^j}\right)
\end{aligned}$$

Since p critical and $df_p = 0$, $\forall l$, $\left.\frac{\partial f}{\partial x^l}\right|_p = 0$, thus the second summation equates to 0 and the transformation law becomes,

$$\left.\frac{\partial^2 f}{\partial u_i \partial u^j}\right|_p = \sum_{k,l}\left(\left.\frac{\partial^2 f}{\partial x^k \partial x^l}\right|_p \left.\frac{\partial x^k}{\partial u^i}\right|_p \left.\frac{\partial x^l}{\partial u^j}\right|_p\right).$$

Notice that we have terms of the Jacobian matrix of the change of coordinates transformation $\vec{x}(\vec{u})$ at p, $J_{ki} = \left.\frac{\partial x^k}{\partial u^i}\right|_p$, thus rewriting the equation above in matrix form,

$$H_{\vec{u}} = J^T H_{\vec{x}} J.$$

This is precisely a congruence relation, and it's precisely how symmetric bilinear forms transform under a change in coordinate/basis. In our case, $H_{\vec{u}}, H_{\vec{x}}$ are coefficient matrices for the same quadratic form under different bases (where the change of basis is given by J.

Finally, taking the determinant of both sides of the equation,

$$\det(H_{\vec{u}}) = \det\left(J^T\right)\det(H_{\vec{x}})\det(J) = \det(J)^2\det(H_{\vec{x}})$$

Since J is invertible, $\det(J) \neq 0$, thus

$$\det(H_{\vec{x}}) = 0 \iff \det(H_{\vec{u}}) = 0,$$

and therefore, non-degeneracy is independent of coordinate change. □

Remark 1.15
Sylvester's Law of Inertia *states, for a real symmetric square matrix A, any congruence transformation of $A \mapsto SAS^T$ for a non-singular matrix S preserves the number of positive and negative eigenvalues ([5, p. 3]).*

In particular, for any symmetric matrix A, there exists a matrix S such that SAS^T is a diagonal matrix D which only has entries $0, +1, -1$. Denote the number of $+1, -1, 0$ entries as n_+, n_-, n_0 respectively. Then,

Definition 1.16 (Index)
The index of a symmetric bilinear matrix A, $\lambda(A) = n_-$ is the maximal dimension of a subspace of V on which H is negative definite, that is, the number of negative eigenvalues.

Definition 1.17 (Nullity)
The nullity of a matrix A, is the dimension of the null-space, thus in the case of a symmetric bilinear matrix, $null(A) = n_0$.

1.3 Vector Fields

A 1-parameter group of diffeomorphisms of a manifold M is a C^∞ map

$$\varphi : \mathbb{R} \times M \longrightarrow M$$

such that

(1) for each $t \in \mathbb{R}$ the map $\varphi_t : M \to M$ defined by

$$\varphi_t(q) = \varphi(t, q)$$

is a diffeomorphism of M onto itself,

(2) for all $t, s \in \mathbb{R}$ we have

$$\varphi_{t+s} = \varphi_t \circ \varphi_s.$$

Given a 1-parameter group φ of diffeomorphisms of M we define a vector field X on M as follows. For every smooth real-valued function f let

$$X_q(f) = \lim_{h \to 0} \frac{f(\varphi_h(q)) - f(q)}{h}.$$

This vector field X is said to *generate* the group φ. Specifically, given a vector field Z, we can define a flow/unique solution of the ODE generated by X as

$$\frac{\mathrm{d}\varphi_t(q)}{\mathrm{d}t} = X_{\varphi_t(q)}, \quad \varphi_0(q) = q,$$

which defines a flow $\{\varphi_t\}_{t \in \mathbb{R}}$ on the manifold M, starting at $\varphi_0(q) = q$. $\{\varphi_t\}_{t \in \mathbb{R}}$ is a group as it satisfies the following,

(1) $\varphi_0 = \mathrm{id}(M)$
(2) $\varphi_{-t} = (\varphi_t)^{-1}$
(3) $\varphi_{t+s} = \varphi_t \circ \varphi_s$

This group structure is important as we want to be able to flow forwards and backwards in t, concatenate flows, and define *deformation homotopies* $t \mapsto \phi_{st}(x)$.

X is an infinitesimal generator of φ in the sense that it describes the instantaneous velocity of the flow at time $t = 0$, for every point $q \in M$ of the manifold,

$$X_q = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \varphi_t(q).$$

If we are evaluating X_q on some function f, then

$$X_q(f) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} f(\varphi_t(q)).$$

Geometrically, this is nothing but the directional derivative of f at q in the direction X_q, which corresponds to the velocity vector of along the curve generated by the flow.

The following theorem is a well known result from differential equations, and will be used in the proofs of the following sections.

Definition 1.18 (Locally Lipschitz in x)
Given some subset $U \in \mathbb{R} \times \mathbb{R}^n$, and a function $F : U \to \mathbb{R}^n$, F is locally lipshitz in x if $\forall (t', x') \in U, \exists V \subseteq U$, V open neighbourhood of (t', x'), and a constant $C > 0$, s.t.

$$\forall (t, x), (s, y) \in V, \|F(t, x) - F(s, y)\| \leq C\|x - y\|$$

Theorem 1.19 (The Picard-Lindelöf Theorem ([3]))
Let U be an open subset of $\mathbb{R} \times \mathbb{R}^n$, $F : U \to \mathbb{R}^n$ a continuous function that's locally Lipschitz in x. Given $(t_0, x_0) \in U$, there exists an $\epsilon > 0$ and a continuously differentiable function $\gamma : (t_0 - \epsilon, t_0 + \epsilon) \to \mathbb{R}^n$ such that for all $t \in (t_0 - \epsilon, t_0 + \epsilon)$, $(t, \gamma(t)) \in U$ and,

$$\frac{\mathrm{d}}{\mathrm{d}t} \gamma(t) = F(t, \gamma(t)).$$

Furthermore, for I_1, I_2 open intervals of $\mathbb{R}$ and continuously differentiable functions $\gamma_i : I_i \to \mathbb{R}^n$ with $(t, \gamma_i(t)) \in U$, and

$$\frac{\mathrm{d}}{\mathrm{d}t} \gamma_i(t) = F(t, \gamma_i(t)),$$

for all $t \in I_i$, if $\exists s \in I_1 \cap I_2$ s.t. $\gamma_1(s) = \gamma_2(s)$, then $\forall t \in I_1 \cap I_2, \gamma_1(s) = \gamma_2(s)$

Lemma 1.20
Let X be a smooth vector field on a manifold M that vanishes outside a compact set $K \subset M$. Then X generates a unique one-parameter group of diffeomorphisms of M.

Proof. Given any smooth curve $\gamma : I \to M$, we define its velocity vector $\dot{\gamma}(t) \in T_{\gamma(t)}M$ as the derivation that maps any arbitrary smooth function f to its rate of change along the curve

$$\dot{\gamma}(t)(f) = \frac{\mathrm{d}\gamma}{\mathrm{d}t}(f) = \lim_{h \to 0} \frac{f(\gamma(t + h)) - f(\gamma(t))}{h}.$$

Let $\{\varphi_t\}$ be the one-parameter group generated by the vector field X. Then for an smooth function f and a point $q \in M$, the curve $t \mapsto \varphi_t(q)$ satisfies $\varphi_0(q) = q$ and

$$\frac{\mathrm{d}\varphi_t(q)}{\mathrm{d}t}(f) = (Xf)_{\varphi_t(q)}.$$

In local coordinates on a chart $U \subset M$, and letting F denote the coordinate representation of X, we obtain a standard ODE,

$$\dot{x}(t) = F(x(t)), \quad x(0) = x_0$$

Since X is smooth, so is F, thus we automatically have F locally lipschitz. By Lemma 1.19, for every initial point x_0, there exists a time interval $(-\epsilon, \epsilon)$ and a unique solution $x(t)$. Therefore, $\forall q \in M, \exists U \subset M$, U containing q, and $\epsilon > 0$ s.t.

$$\frac{\mathrm{d}\varphi_t(q)}{\mathrm{d}t} = X_{\varphi_t(q)}, \quad \varphi_0(q) = q$$

has a unique solution for $|t| < \epsilon$.

Since the compact set $K \subset M$ can be covered by finitely many such coordinate neighborhoods $U_{p_1}, \ldots, U_{p_N}$, let $\epsilon_0 = \min\{\epsilon_{p_1}, \ldots, \epsilon_{p_N}\} > 0$. Then for every $q \in K$, there exists an index i such that $q \in U_{p_i}$, and hence a unique solution exists on the uniform time interval $(-\epsilon_0, \epsilon_0)$. If two local solutions are defined on overlapping neighborhoods, they agree on the overlap by the uniqueness part of the local existence theorem, since they solve the same ODE with the same initial condition. Therefore, the locally defined solutions glue together to produce a single well-defined solution on $(-\epsilon_0, \epsilon_0)$ for all initial points in K.

Set $\varphi_t(q) = q$ for all $q \notin K$. Because X vanishes outside K, the solution extends uniquely for all $|t| < \epsilon_0$ and for all $q \in M$. Hence the flow φ_t is globally defined and smooth in both t and q. It also satisfies $\varphi_{t+s} = \varphi_t \circ \varphi_s$ for $|t|, |s|, |t+s| < \epsilon_0$.

For $t \geq \epsilon_0$, any such t can be expressed as $t = k\left(\frac{\epsilon_0}{2}\right) + r$ for $k > 0$ and $|r| < \frac{\epsilon_0}{2}$, then we can decompose φ_t as,

$$\varphi_t = \varphi_{\epsilon_0} \circ \cdots \circ \varphi_{\epsilon_0} \circ \varphi_r$$

Thus the solution is defined for all $t \in \mathbb{R}$.

Consequently, the vector field X generates a unique global flow, i.e. a one-parameter group of diffeomorphisms of M. $\square$

2 Morse Functions and the Morse Lemma

With the foundational differential-geometric machinery established, we now introduce the central object of study. A Morse function is a smooth function that is as generic as possible, one with no degenerate critical points, so that the local behavior near each critical point is completely controlled by the Hessian. Before developing the theory further, we highlight a few interesting and important properties of Morse functions ([4, Ch. 1.1]).

Definition 2.1 (Morse Functions)
A smooth map $f : M \to \mathbb{R}$ on a manifold M is a Morse function if all of its critical points are non-degenerate.

Theorem 2.2 (Every manifold has a Morse function)
On a compact, smooth n-manifold M, any continuous function can be approximated by a Morse function.

Corollary 2.3
Thm 2.2 implies that the set of Morse functions of a compact smooth n-manifold M forms an open and dense subset of smooth functions in C^2 topology.

Operating in the C^2 topology ensures that perturbations of the smooth functions do not affect the second derivative values in an uncontrollable manner, as we want to ensure the Hessian remains non-0.

Definition 2.4 (Self-Indexing Morse Functions)
A Morse function f is self-indexing if $f(p) = \lambda(p)$ for all critical points p.

Theorem 2.5 (Every manifold has a self-indexing Morse function)
Let M be a smooth, closed, connected n-dimensional manifold, then M admits a self-indexing Morse function with a unique local minimum and maximum (one critical point of index 0 and one critical point of index n).

The Morse Lemma gives a canonical local normal form for a Morse function near each critical point. Central to its proof is the following decomposition lemma, which expresses any smooth function in a neighborhood of the origin as a weighted sum of coordinate functions.

Lemma 2.6
Let f be a C^∞ function in a convex neighborhood V of 0 in $\mathbb{R}^n$. We can assume W.L.O.G that $f(0) = 0$, as constant shifts vanish under differentiation. Then,

$$f(x_1, \ldots, x_n) = \sum_{i=1}^{n} x_i g_i(x_1, \ldots, x_n)$$

for some suitable C^∞ functions g_i defined in V, with $g_i(0) = \frac{\partial f}{\partial x_i}(0)$.

Proof. Since V is convex and contains 0, for any $t \in [0, 1]$, the entire line $t \mapsto (tx_1, \ldots, tx_n)$ is also contained in V.

Let $\phi(t) = f(tx_1, \ldots, tx_n)$, we have $\phi(0) = f(0) = 0, \phi(1) = f(x_1, \ldots, x_n)$, thus by the fundamental theorem of calculus and another application of the chain rule,

$$\begin{aligned}
f(x_1,\dots,x_n) &= \phi(1)-\phi(0)\\
&= \int_0^1 \frac{\mathrm{d}\phi}{\mathrm{d}t}\\
&= \int_0^1 \frac{\mathrm{d}f(tx_1,\dots,tx_n)}{\mathrm{d}t}\,\mathrm{d}t\\
&= \int_0^1 \sum_{i=1}^n \frac{\partial f(tx_1,\dots,tx_n)}{\partial x_i}\frac{\partial(tx_i)}{\partial t}\,\mathrm{d}t\\
&= \int_0^1 \sum_{i=1}^n \frac{\partial f(tx_1,\dots,tx_n)}{\partial x_i}\,x_i\,\mathrm{d}t
\end{aligned}$$

Note the distinction between x_i's the partial derivatives $\frac{\partial}{\partial x_i}$ versus the coefficient term. The former is the i-the partial derivative of f as a function on $\mathbb{R}^n$, whereas the latter is a constant that represents the i-th coordinate of the input $(x_1,\dots,x_n)$.

Therefore, we can define

$$g_i(x_1,\dots,x_n) := \int_0^1 \frac{\partial f(tx_1,\dots,tx_n)}{\partial x_i}\,\mathrm{d}t.$$

It is C^∞ and is defined in V by the same properties of f, and $g_i(0)=\int_0^1 \frac{\partial f(0)}{\partial x_i}\,\mathrm{d}t = 0$. □

Theorem 2.7 (Morse Lemma)
Let f be a smooth function on M and let p be a non-degenerate critical point for f. Then there exists a local coordinate system $(x^1,\dots,x^n)$ in a neighborhood U of p s.t. $x^i(p)=0$, and

$$f = f(p)-(x^1)^2-\cdots-(x^\lambda)^2+(x^{\lambda+1})^2+\cdots+(x^n)^2$$

holds throughout U, where λ is the index of f at p.

Proof. We first argue if there exists such an expression for f, then λ has to be the index of f at p. For any coordinates system, if $(z^1,\dots,z^n)$ if

$$f(q) = f(p)-(z^1(q))^2-\cdots-(z^\lambda(q))^2+(z^{\lambda+1}(q))^2+\cdots+(z^n(q))^2,$$

we must have,

$$\frac{\partial^2 f}{\partial x^i\partial x^j}(p) = \begin{cases} -2 & \text{if } i=j\le\lambda,\\ 2 & \text{if } i=j>\lambda,\\ 0 & \text{otherwise.}\end{cases}$$

where the matrix representing f_{**} is,

$$\begin{pmatrix} -2 & & & & & \\ & \ddots & & & & \\ & & -2 & & & \\ & & & 2 & & \\ & & & & \ddots & \\ & & & & & 2 \end{pmatrix}.$$

This implies that there exists a subspace $E \subset T_pM$ with $\dim(E) = \lambda$ on which f_{**} is negative definite, and a subspace $V \subset T_pM$ with $\dim(V) = n - \lambda$ on which f_{**} is positive definite. If there existed a subspace of dimension $> \lambda$ on which f_{**} were negative definite, then it would intersect V, which is impossible.

Therefore, λ is the index of f_{**}.

Now we want to show such a coordinate system exists. Let p be a nondegenerate critical point of a smooth function f. We first translate coordinates so that p becomes the origin and

$$f(p) = f(0) = 0.$$

By Lemma 2.6, we can write f near 0 as

$$f(u_1, \ldots, u_n) = \sum_j u_j\, g_j(u_1, \ldots, u_n),$$

for $(u_1, \ldots, u_n)$ in some neighborhood of 0 and $g_j \in C^\infty$. Furthermore, $g_j(0) = \frac{\partial f}{\partial u_j}(0) = 0$ as 0 is a critical point of f. This enables us to further apply Lemma 2.6 to g_j,

$$g_j(u_1, \ldots, u_n) = \sum_i u_i\, H_{ij}(u_1, \ldots, u_n),$$

for $h_{ij} \in C^\infty$. Combining the two above, we have that

$$f(u_1, \ldots, u_n) = \sum_{i,j} u_i u_j\, H_{ij}(u_1, \ldots, u_n).$$

We may assume the matrix (H_{ij}) is symmetric by replacing

$$H_{ij} \leftarrow \tfrac{1}{2}(H_{ij} + H_{ji}),$$

since $u_i u_j H_{ij} + u_j u_i H_{ji}$ contributes symmetrically to the same quadratic term.

At the origin,

$$(H_{ij}(0)) = \mathrm{Hess}_0(f),$$

the Hessian matrix of f at 0. Since the critical point is nondegenerate, this matrix is non-singular.

We now imitate the usual diagonalization process for quadratic forms. The proof proceeds by induction on the number of coordinates.

Assume that after a linear change of coordinates we have already "peeled off" $r-1$ variables, so that f can be written as

$$f(v_1, \ldots, v_n) = \pm v_1^2 \pm \cdots \pm v_{r-1}^2 + \sum_{i,j \geq r} v_i v_j H_{ij}(v_1, \ldots, v_n),$$

where (H_{ij}) is symmetric and smooth.

Lemma 2.8
By performing a linear change among the variables $v_r, \ldots, v_n$, we may assume $H_{rr}(0) \neq 0$.

Proof of Lemma 2.8. To be more precise, consider the following Hessian matrix (already evaluated at $p = 0$),

$$H = \begin{pmatrix} \pm 1 & & & & & & \\ & \pm 1 & & & & & \\ & & \ddots & & & & \\ & & & \pm 1 & & & \\ & & & & h_{rr} & \cdots & h_{rn} \\ & & & & h_{(r+1)r} & \cdots & h_{(r+1)n} \\ & & & & \vdots & \ddots & \vdots \\ & & & & h_{nr} & \cdots & h_{nn} \end{pmatrix}.$$

If $h_{rr} \neq 0$, then we are done. Otherwise, if $h_{rr} = 0$, then at least one of the entries h_{rj} for $(r+1) \leq j \leq n$ must be nonzero. Indeed, if all such entries were zero, then the r-th row (and column) would vanish in the remaining block, implying that the Hessian matrix is not full rank. This contradicts the non-degeneracy of the critical point p.

Now consider the coordinate change

$$y_r = v_r + t v_j, \qquad y_i = v_i \quad \text{for } i \neq r.$$

This transformation is represented by the change-of-basis matrix $P = I + tE_{jr}$. Under this change of coordinates the Hessian transforms as $H' = P^T H P$.

We now compute the (r, r) entry of the new matrix. By definition,

$$H'_{rr} = e_r^T (P^T H P) e_r = (P e_r)^T H (P e_r).$$

Since $Pe_r = e_r + te_j$, we obtain

$$H'_{rr} = (e_r + te_j)^T H (e_r + te_j).$$

Expanding this expression gives

$$H'_{rr} = h_{rr} + 2t h_{rj} + t^2 h_{jj}.$$

Because we assumed $h_{rr} = 0$, this simplifies to

$$H'_{rr}(t) = t(2h_{rj} + t h_{jj}).$$

Since $h_{rj} \neq 0$, and $H'_{rr}(t)$ is a quadratic polynomial that's not identically 0, with two real roots $t = 0, -\frac{2h_{rj}}{h_{jj}}$. Therefore, we can choose t s.t. $t \neq 0, -\frac{2h_{rj}}{h_{jj}}$, which guarantees that $H'_{rr} \neq 0$, as desired. □

Define

$$g(v_1, \ldots, v_n) = \sqrt{|H_{rr}(v_1, \ldots, v_n)|}.$$

Since $H_{rr}(0) \neq 0$, there exists a neighborhood of 0 on which H doesn't vanish, thus g is also non-vanishing, and smooth. Now introduce new coordinates

$$\begin{aligned} V_i &= v_i, \qquad i \neq r, \\ V_r &= g(v_1, \ldots, v_n) \left(v_r + \sum_{i > r} v_i H_{ir}(v_1, \ldots, v_n) \right). \end{aligned}$$

Since g and H are smooth and non vanishing, this change of variables is smooth and locally invertible near the origin by the inverse function theorem.

Now isolate the terms of f involving v_r:

$$\sum_{i,j\geq r} v_i v_j H_{ij} = H_{rr} v_r^2 + 2\sum_{i>r} v_i v_r H_{ir} + \sum_{i,j>r} v_i v_j H_{ij}.$$

Completing the square in v_r gives

$$H_{rr}\left(v_r + \sum_{i>r} \frac{H_{ir}}{H_{rr}} v_i\right)^2 + \sum_{i,j>r} v_i v_j \left(H_{ij} - \frac{H_{ir}H_{jr}}{H_{rr}}\right).$$

Using the coordinate V_r defined above, the first term becomes

$$\operatorname{sign}(H_{rr}) V_r^2.$$

Hence the function takes the form

$$f = \pm v_1^2 \pm \cdots \pm v_{r-1}^2 + \operatorname{sign}(H_{rr}) V_r^2 + \sum_{i,j>r} V_i V_j \widetilde{H}_{ij}(V),$$

where

$$\widetilde{H}_{ij} = H_{ij} - \frac{H_{ir}H_{jr}}{H_{rr}}.$$

Thus we have extracted one more pure quadratic term. This completes the inductive step.

Continuing inductively, we obtain coordinates $(x_1, \ldots, x_n)$ in which

$$f(x) = -x_1^2 - \cdots - x_\lambda^2 + x_{\lambda+1}^2 + \cdots + x_n^2.$$

where λ is the index of the critical point. □

As a consequence of the Morse Lemma, we have the following corollary.

Corollary 2.9
Non-degenerate critical points are isolated.

3 Interval sublevel set containing no critical points

Let $M^t := f^{-1}((-\infty, t])$ denote the sublevel set of M containing the preimage of values less than t. For some $a < b$, this section explores the changes to M^t as t goes from a to b when the interval sublevel set $f^{-1}([a,b])$ contains no critical points of f.

We first establish some definitions and concepts.

Definition 3.1 (Riemannian Metric)
For a manifold M, a Riemannian metric assigns to each point $p \in M$ a symmetric positive-definite bilinear form g_p s.t.

(1) $g_p(u,v) = g_p(v,u) \quad \forall u, v \in T_pM$
(2) $g_p(u,u) > 0 \quad \forall u \neq 0$

and g_p varies smoothly in p.

The Riemannian metric allows us to measure lengths, angles , and distances. Each g_p is itself an inner product on T_pM.

Theorem 3.2 (Partition of Unity)
For any open cover $\{U_\alpha\}_{\alpha \in I}$ of a space X, there exists a partition (family of continuous functions $X \to [0,1]$) $\{\phi_\alpha\}_{\alpha \in I}$ indexed over the same I, s.t.

(1) $0 \leq \phi_\alpha \leq 1$
(2) $\operatorname{supp}(\phi_\alpha) \subset U_\alpha$, *where* $\operatorname{supp}(\phi_\alpha) := \overline{\{x \in X | \phi_\alpha(x) \neq 0\}}$
(3) $\{\phi_\alpha\}_{\alpha \in I}$ *is locally finite, that is, $\forall x \in X, \exists$ neighborhood W of p s.t. $\phi_\alpha\Big|_W \equiv 0$ for all but finitely many α*
(4) $\sum_\alpha \phi_\alpha = 1$

Note that the third condition in Theorem 3.2 ensures that the sum $\sum_\alpha \phi_\alpha$ is a finite sum, hence preserves the continuity conditions of the individual ϕ_α functions. Furthermore, if X is a smooth manifold, we will also have smooth partitions.

Theorem 3.3
Let $f : M \to \mathbb{R}$ be a smooth function, and let $a < b$ be such that $f^{-1}[a,b]$ is compact and contains no critical points of f. Then the sublevel sets $M^a \simeq M^b$ are diffeomorphic. In particular, M_a is a deformation retract of M_b, and the inclusion $M_a \hookrightarrow M_b$ is a homotopy equivalence.

Proof. Choose a Riemannian metric g on M. Let $\langle X, Y \rangle$ denote the inner product of tangent vectors determined by g, that is, $\langle X, Y \rangle = g(X,Y)$.

We define the gradient of f, $\operatorname{grad} f$ through the relation

$$\langle \operatorname{grad} f, X \rangle = X(f) = df(X)$$

for all vector fields X. Note that df is the exterior derivative of f (Def 1.6). The gradient varies smoothly and vanishes precisely at the critical points of f, as the partial derivatives in all directions are 0.

If $\gamma : \mathbb{R} \to M$ is a curve, then

$$\langle \frac{\mathrm{d}\gamma}{\mathrm{d}t}, \operatorname{grad} f \rangle = \frac{\mathrm{d}(f \circ \gamma)}{\mathrm{d}t}.$$

Since $f^{-1}[a,b]$ is compact, we can choose an open neighborhood $U \subset M$ s.t. U *does not* contain a critical point, $f^{-1}[a,b] \subset U$, and $\overline{U}$ compact. $V := M \setminus f^{-1}[a,b]$ is also an open set, and $\{U, V\}$ is an open cover of M. On U, define a smooth function $\rho : M \to \mathbb{R}$ by

$$\rho(q) = \frac{1}{\langle \operatorname{grad} f, \operatorname{grad} f \rangle_q}.$$

Define a vector field X on U by

$$X_q = \rho(q)(\operatorname{grad} f)_q.$$

On V define the zero vector field, 0.

Now, the Partition of Unity Theorem (Theorem 3.2) guarantees the existence of smooth functions ϕ_U, ϕ_V s.t. $0 \leq \phi_U, \phi_V \leq 1$, $\operatorname{supp}(\phi_U) \subset U$, $\operatorname{supp}(\phi_V) \subset V$, and $\phi_U + \phi_V = 1$. Now we can define a global vector field on M as follows,

$$\tilde{X} := \phi_U X + \phi_V 0 = \phi_U X.$$

This is well defined as multiplying a smooth function by smooth vector field gives a smooth vector field, and the sum of smooth vector fields is also smooth. Since $f^{-1}[a,b] \subset U$ and $\phi_U \equiv 1$ near $f^{-1}[a,b]$, we have that $\tilde{X} = X$ on $f^{-1}[a,b]$. Outside this compact support U, $\phi_U = 0$ thus $\tilde{X} = 0$. Therefore, $\tilde{X}$ vanishes outside U, and is compactly supported in U.

Then, on $f^{-1}[a,b]$,

$$\begin{aligned}\langle X, \operatorname{grad} f \rangle &= \langle \rho \operatorname{grad} f, \operatorname{grad} f \rangle \\ &= \rho \langle \operatorname{grad} f, \operatorname{grad} f \rangle \\ &= 1.\end{aligned}$$

Therefore, X satisfies the hypotheses of Lemma 1.20. Let $\{\varphi_t\}$ be the flow generated by X. Then along any flow line,

$$\begin{aligned}df(\varphi_t(q)) &= \langle d\varphi_t(q), \operatorname{grad} f \rangle \\ &= \langle X, \operatorname{grad} f \rangle \\ &= 1\end{aligned}$$

as long as $\varphi_t(q) \in f^{-1}[a,b]$. Hence the flow $t \mapsto f(\varphi_t(q))$ moves points monotonically at a linear rate in the direction of increasing f without leaving the sublevel set. Thus $\varphi_{b-a} : M \to M$ defines a diffeomorphism from M_a onto M_b.

Define a one-parameter family of maps as follows,

$$r_t : M^b \to M^b$$

$$r_t(q) = \begin{cases} q, & \text{if } f(q) \leq a, \\ \varphi_{t(a-f(q))}(q), & \text{if } a \leq f(q) \leq b. \end{cases}$$

Then $r_0 = \mathrm{id}$, and$r_1(q) = \varphi_{a-f(q)}(q)$, which is a retraction from M^b onto M^a. Hence M^a is a deformation retract of M^b, and they are homotopy equivalent. □

4 Interval sublevel set containing a non-degenerate critical point

In the previous section, we showed that there are no topological changes between interval sublevel sets that do not contain a critical point. The more interesting case is looking at what happens when the interval sublevel set $f^{-1}([a,b])$ do contain a non-degenerate critical point.

Theorem 4.1
Let $f : M \to \mathbb{R}$ be a smooth function. Let p be a non-degenerate critical point of f with index λ. Set $f(p) = c$ and suppose $f^{-1}[c-\epsilon, c+\epsilon]$ compact and contains no critical point of f other than p, for some sufficiently small $\epsilon > 0$. Then $M^{c+\epsilon}$ has homotopy type $M^{c-\epsilon}$ with a λ cell e^λ attached.

Proof of Theorem 4.1. The proof involves defining an auxiliary function based on f that "pushes" down the values of f in between the sublevel sets. A map of the proof is provided in Fig 1.

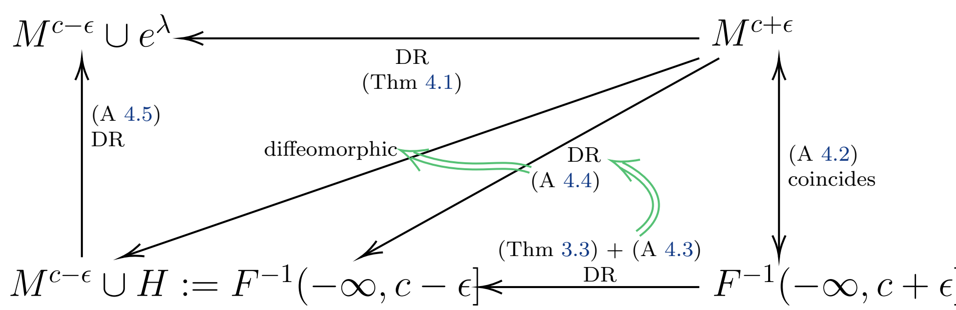


Figure 1: Map of Proof of Theorem 4.1

By the Morse Lemma (Thm 2.7), there exists a local coordinate system $u^1, \dots, u^n$ of a neighborhood U of p such that,

$$f = c - (u^1)^2 - \cdots - (u^\lambda)^2 + (u^{\lambda+1})^2 + \cdots + (u^n)^2$$

holds throughout U. In particular, $u^i(p) = 0, \forall i$. Choose ϵ sufficiently small so that

(1) $f^{-1}[c-\epsilon, c+\epsilon]$ compact and contains no critical point of f other than p
(2) The image of U under the diffeomorphic embedding

$$(u_1, \dots, u^n) : U \to \mathbb{R}^n$$

contains the closed ball $B_{2\epsilon} = \{(u_1, \dots, u^n) \mid \sum (u^i)^2 \leq 2\epsilon\}$.

Define

$$e^\lambda := \{(u_1, \dots, u^n) \mid (u^1)^2 + \cdots + (u^\lambda)^2 \leq \epsilon \wedge (u^{\lambda+1})^2 = \cdots = (u^n)^2 = 0\}.$$

In particular, $e^\lambda \cap M^{c-\epsilon} = \partial e^\lambda$ indicates to that e^λ is attached to $M^{c-\epsilon}$ as desired. A diagrammatic picture of our set up is shown in Fig 2.

Let $\mu : \mathbb{R} \to \mathbb{R}$ be a smooth function satisfying the following conditions,

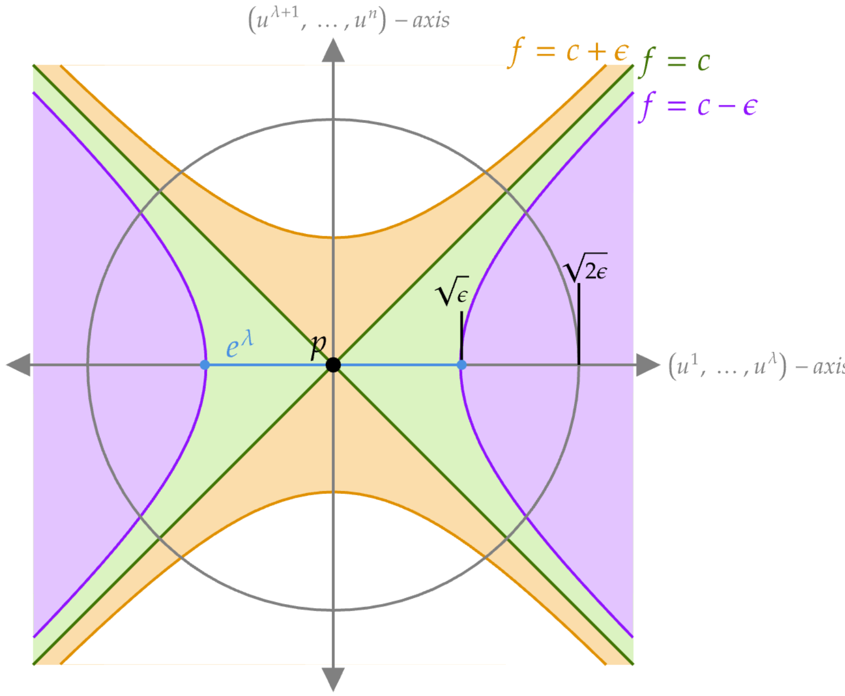


Figure 2: 2D projection of of the level sets of f at $c+\epsilon$ (orange), c (green), $c-\epsilon$(purple) onto the plane with the $(u^1, \ldots, u^\lambda)$-axis and the $(u^{\lambda+1}, \ldots, u^n)$-axis. The grey circle denotes the closed ball $B_{2\epsilon}$ The critical point p is the origin of this axes, and the e^λ-cell is the blue line.

(1) $\mu(0) > \epsilon$
(2) $\mu(r) = 0$ for $r > 2\epsilon$
(3) $\forall r, -1 < \mu'(r) \leq 0$

Thus μ is a function that looks like the graph in Fig 3.

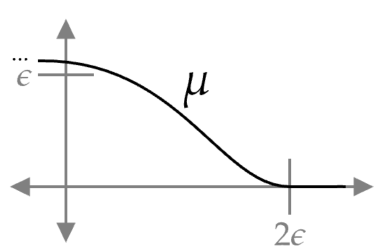


Figure 3: Approximate graph of μ

Furthermore, define $\xi, \eta : U \to [0, \infty)$ with

$$\xi = (u^1)^2 + \cdots + (u^\lambda)^2 \tag{1}$$
$$\eta = (u^{\lambda+1})^2 + \cdots + (u^n)^2 \tag{2}$$

. Then $f = c - \xi + \eta$.

We now construct a smooth function $F : M \to \mathbb{R}$ as

$$F = f - \mu((u^1)^2 + \cdots + (u^\lambda)^2 + 2(u^{\lambda+1})^2 + \cdots + 2(u^n)^2),$$

thus for $q \in U$,

$$F(q) = c - \xi(q) + \eta(q) - \mu(\xi(q) + 2\eta(q))$$

Assertion 4.2
$F^{-1}(-\infty, c+\epsilon]$ coincides with $M^{c+\epsilon}$

Outside of the ellipsoid $\xi + 2\eta \leq 2\epsilon$, $F = f$ since $\mu(\xi + 2\eta) = 0 \implies F = f - 0 = f$. Since $F(r) > 0$ for $r \in [0, 2\epsilon]$ and $\xi + 2\eta \leq 2\epsilon \implies \frac{1}{2}\xi + \eta \leq \epsilon$, within the ellipsoid,

$$\begin{aligned} F \leq f &= c - \xi + \eta \\ &\leq c - \frac{1}{2}\xi + \eta \\ &\leq c + \frac{1}{2}\xi + \eta \\ &\leq c + \epsilon. \end{aligned}$$

This shows that F decreases from the value of f in a smooth way within the ellipsoid and $F \leq f$. These two cases show that the sublevel sets $F^{-1}(-\infty, c+\epsilon]$ and $M^{c+\epsilon}$ coincide on the level of sets, thus proving Assertion 4.2.

Assertion 4.3
Critical points of F are the same as that of f. In particular, $F^{-1}[c-\epsilon,\, c+\epsilon]$ has no critical points of F.

Note that

$$\begin{aligned} \frac{\partial F}{\partial \xi} &= -1 - \mu'(\xi + 2\eta) < 0, \\ \frac{\partial F}{\partial \eta} &= 1 - 2\mu'(\xi + 2\eta) \geq 1. \end{aligned}$$

Since

$$dF = \frac{\partial F}{\partial \xi}\, d\xi + \frac{\partial F}{\partial \eta}\, d\eta,$$

and $d\xi$ and $d\eta$ vanish simultaneously only at the origin, it follows that F has no critical points in U other than at the origin. By Assertion 4.2, we have

$$F^{-1}[c-\epsilon,\, c+\epsilon] \subset f^{-1}[c-\epsilon,\, c+\epsilon].$$

Hence $F^{-1}[c-\epsilon,\, c+\epsilon]$ is compact and contains no critical points of F except possibly at p. However,

$$F(p) = c - \mu(0) < c - \epsilon,$$

which means $p \notin F^{-1}[c-\epsilon,\, c+\epsilon]$ Therefore $F^{-1}[c-\epsilon,\, c+\epsilon]$ contains no critical points.

Assertion 4.4
$F^{-1}(-\infty, c-\epsilon]$ is a deformation retract of $M^{c+\epsilon}$.

Since $F^{-1}[c-\epsilon,\, c+\epsilon]$ contains no critical points, $F^{-1}(-\infty,\, c-\epsilon]$ is a deformation retract of and is diffeomorphic to $F^{-1}(-\infty,\, c+\epsilon] = M^{c-\epsilon}$ by Thm 3.3. We can denote $F^{-1}(-\infty, c-\epsilon]$ as $M^{c-\epsilon} \cup H$, where $H = F^{-1}(-\infty, c-\epsilon] - M^{c-\epsilon}$ is called a Handle. We define the notion of a Handle in Def 5.4.

Note that e^{λ} is contained in the handle H, which is diffeomorphic to $D^{\lambda} \times D^{n-\lambda}$, with $e^{\lambda} = D^{\lambda} \times \{0\}$. For $q \in e^{\lambda}$, $F(q) \leq F(p) < c-\epsilon$ since $\frac{\partial F}{\partial \xi} < 0$, however, $f(q) \geq c-\epsilon$. This illustrates that the e^{λ}-cell is contained in the sublevel set $F^{-1}(-\infty, c-\epsilon]$, we have verified that it is indeed contained in H, however it is not already contained in the sublevel set $M^{c-\epsilon}$.

We can plot the sublevel sets of F as in Fig 4.

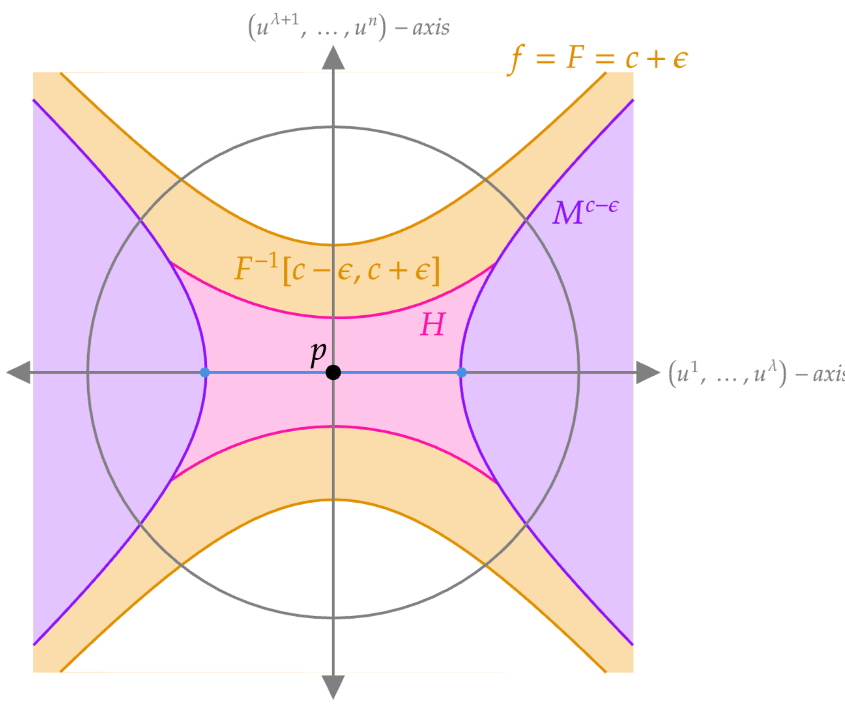


Figure 4: Sublevel sets of F, where the purple area is the component of $F^{-1}(-\infty, c-\epsilon]$ that coincides with $M^{c-\epsilon}$, whereas H is shaded pink.

Assertion 4.5
$M^{c-\epsilon} \cup e^{\lambda}$ is a deformation retract of $M^{c-\epsilon} \cup H$

Define a retraction

$$r_t\colon M^{c-\epsilon} \cup H \longrightarrow M^{c-\epsilon} \cup H,$$

where r_t is the identity outside of U. Within U, we have three cases corresponding to the three regions shown in Fig 5.

(1) Within $\xi \leq \epsilon$, let r_t map

$$(u^1, \dots, u^n) \longmapsto (u^1, \dots, u^{\lambda},\, tu^{\lambda+1}, \dots, tu^n).$$

Thus $r_1 = \text{id}$, and r_0 maps the strip into e^{λ}.

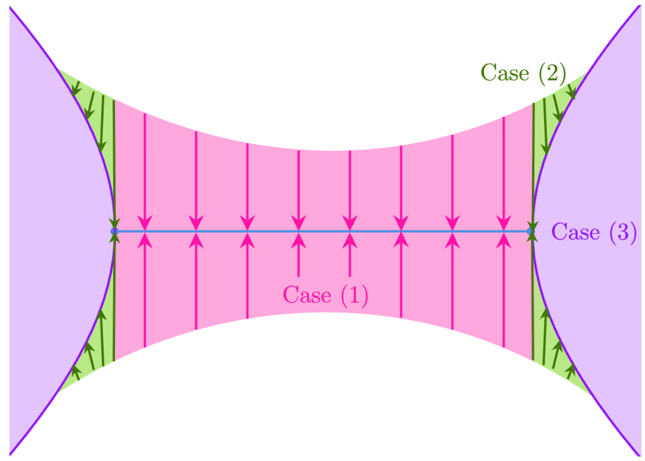


Figure 5: Piecewise deformation retract of $M^{c-\epsilon} \cup H$ to $M^{c-\epsilon} \cup e^\lambda$. For $u \in U$, Case (1) corresponds to $\xi(u) \leq \epsilon$, case (2) to $\epsilon \leq \xi(u) \leq \eta(u) + \epsilon$, and case (3) to $\eta(u) + \epsilon \leq \xi(u)$.

(2) Within $\epsilon \leq \xi \leq \eta + \epsilon$, let r_t correspond to

$$(u^1, \ldots, u^n) \longmapsto (u^1, \ldots, u^\lambda,\, s_t u^{\lambda+1}, \ldots, s_t u^n),$$

where

$$s_t = t + (1-t)\left(\frac{\xi - \epsilon}{\eta}\right)^2 \in [0, 1].$$

$\left(\frac{\xi-\epsilon}{\eta}\right)$ normalizes our interval $\xi \in [\epsilon, \epsilon + \eta]$ to be $[0, 1]$, the squaring is an arbitrary choice to "smoothen" out the retraction. Thus $r_1 = \mathrm{id}$, and r_0 maps the set of points into $f^{-1}(c - \epsilon)$. This coincides with case (1) when $\xi = \epsilon$.

(3) Within $\eta + \epsilon \leq \xi$ (that is they lie in $M^{c-\epsilon}$), let r_t be the identity. This coincides with case (2) when $\xi = \eta + \epsilon$

Therefore $M^{c-\epsilon} \cup e^\lambda$ is a deformation retract of $M^{c-\epsilon} \cup H$ and thus a deformation retract of $F^{-1}(-\infty,\, c + \epsilon]$. This concludes the proof of Thm 4.1. $\square$

5 CW-complex and Handlebody decomposition

The aim of this section is to make precise the sense in which the critical point data of a Morse function completely determines the topology of the underlying manifold. We prove that a compact manifold M has the homotopy type of a CW-complex, then pass to the smooth setting by replacing cells with handles, to obtain a handlebody decomposition of M that is both geometrically explicit and entirely dictated by the index data of the Morse function.

Definition 5.1 (CW-complex)
A CW-complex is the union

$$X = \bigcup X_k$$

of a sequence of topological spaces

$$\emptyset = X_0 \subset X_1 \subset X_2 \subset \cdots$$

such that X_k is obtained from X_{k-1} by gluing copies of k-cells (e^k_α) to X_{k-1} with gluing maps $g_\alpha : \partial e^k \to X_{k-1}$. Thus, $X_k = X_{nk1} \cup e^k_\alpha$.

Remark 5.2
Each X_k is called the k-skeleton of the CW complex $X = \bigcup X_k$.

Remark 5.3 (CW stands for "closure finite" and "weak")
For every cell in a CW complex $X = \bigcup X_k$, its closure must intersect only a finite number of other cells. Furthermore, X is equipped with the weak topology, which means $U \subset X$ is open $\Leftrightarrow U \cap X_k$ open for each k.

Definition 5.4 (Handlebody)
Let X be a smooth n-dimensional manifold with boundary ∂X. For $0 \leq k \leq n$, an n-dim k-handle H is a contractable smooth manifold

$$D^k \times D^{n-k} \simeq D^n$$

attached to ∂X along its attaching region,

$$\partial D^k \times D^{n-k},$$

with a smooth attaching map $\phi : \partial D^k \times D^{n-k} \to \partial X$.

After a smoothing of corners, $X \cup_\phi H$ is a smooth n-manifold with boundary. We further define the attaching sphere $A^k := \partial D^k \times \{0\}$, the core $C^k := D^k \times \{0\}$, the belt sphere $B^{n-k} := \{0\} \times \partial D^{n-k}$ and the co-core $K^{n-k} := \{0\} \times D^{n-k}$.

As an example, Fig 6 illustrates a 3-dim 2-handle, $D^2 \times D^1$, and a 3-dim 1-handle, $D^1 \times D^2$, and how they may attach onto a 3-manifold with boundary.

A k-handle, $D^k \times D^{n-k}$, can be viewed as a thickened version of a k-cell D^k, where the extra D^{n-k} factor ensures that the resulting space has dimension n. In this sense, a handle decomposition is a "thickened" analogue of a CW-complex. A handle decomposition of an n-manifold is constructed by attaching k-handles along their attaching regions $\partial D^k \times D^{n-k}$ to the boundary of a previously built n-manifold, starting from a 0-handle. By contrast, a CW-complex is built by attaching k-cells D^k to the $(k-1)$-skeleton via maps from ∂D^k.

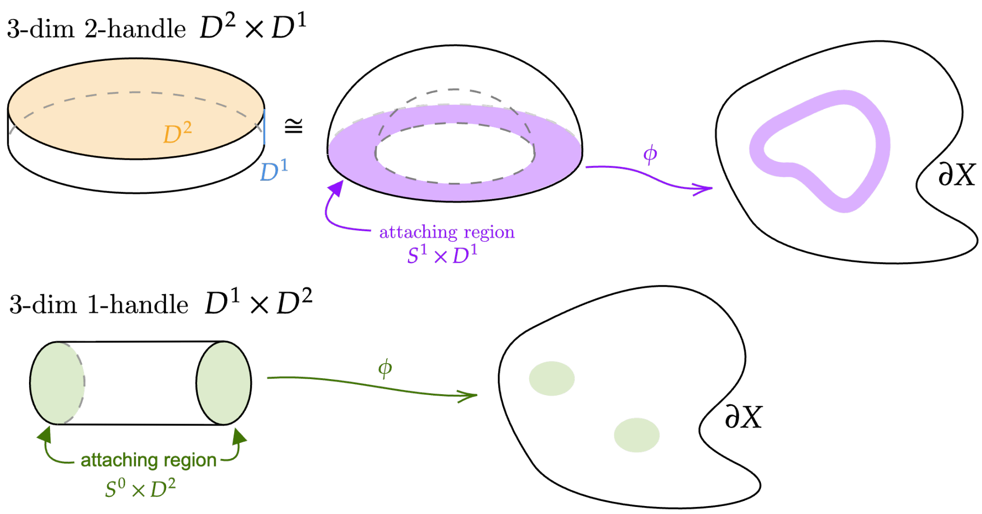


Figure 6: Embedding of 3-dim 2- and 1-handles via attaching maps ϕ into ∂X, where $dim(\partial X) = 2$.

While CW-complexes are extremely flexible and useful for studying homotopy type, they do not in general preserve manifold structure. Indeed, most CW-complexes are not manifolds. On the other hand, handle decompositions, are specifically designed to respect smooth manifold structure, allowing one to build manifolds inductively while keeping track of their topology and geometry.

Lemma 5.5 (Whitehead)
Let φ_0 and φ_1 be homotopic maps from the sphere ∂e^λ to X. Then the identity map of X extends to a homotopy equivalence

$$K : X \cup_{\varphi_0} e^\lambda \longrightarrow X \cup_{\varphi_1} e^\lambda$$

Proof. Define K by the formulas,

$$\begin{aligned} K(x) &= x && \text{for } x \in X, \\ K(tu) &= 2tu && \text{for } 0 \le t \le \tfrac{1}{2},\ u \in \partial e^\lambda, \\ K(tu) &= \varphi_{2-2t}(u) && \text{for } \tfrac{1}{2} \le t \le 1,\ u \in \partial e^\lambda. \end{aligned}$$

Here φ_t denotes the homotopy between φ_0 and φ_1, and tu denotes the product of the scalar t with the unit vector u.

We show that this is a well-defined map between $X \cup_{\varphi_0} e^\lambda$ and $X \cup_{\varphi_1} e^\lambda$. Note that $X \cup_{\varphi_0} e^\lambda$ is defined as $(X \sqcup D^\lambda)/\{u \sim \varphi_0(u) \mid u \in \partial e^\lambda\}$. Therefore for a map to be continuous on $X \cup_{\varphi_0} e^\lambda$, we must have that it maps u and $\varphi_0(u)$ to the same thing for $u \in \partial e^\lambda$. By the first condition of K, $K(u) = u$ since $u \in X$ as well, by the third condition, for $t = 1$, $K(1u) = \varphi_{2-2}(u) = \varphi_0(u) = u$. Thus $K(u) = K(\varphi_0(u))$. Therefore, this map is well defined.

Furthermore, for the map to be continuous, it also has to preserve the quotient conditions of $X \cup_{\varphi_1} e^\lambda = (X \sqcup D^\lambda)/\{u \sim \varphi_1(u) \mid u \in \partial e^\lambda\}$. For $v \sim \varphi_1(v) \in \partial e^\lambda$, we have that $K(\frac{1}{2}v) = 2\frac{1}{2}v = v$ by the second condition, and $K(\frac{1}{2}v) = \varphi_{2-2(\frac{1}{2})}(v) = \varphi_1(v)$ by the third condition. Thus $v \sim \varphi_1(v)$.

A corresponding map

$$\ell : X \cup_{\varphi_1} e^\lambda \longrightarrow X \cup_{\varphi_0} e^\lambda$$

is defined by similar formulas, in particular,

$$\begin{aligned} \ell(x) &= x && \text{for } x \in X, \\ \ell(tu) &= 2tu && \text{for } 0 \le t \le \tfrac{1}{2},\ u \in \partial e^\lambda, \\ \ell(tu) &= \varphi_{2t-1}(u) && \text{for } \tfrac{1}{2} \le t \le 1,\ u \in \partial e^\lambda. \end{aligned}$$

We now verify ℓK and $K\ell$ are homotopic to the respective identity maps. Computing lK explicitly, for $x \in X$, $\quad lK(x) = l(x) = x$, and for $u \in \partial e^\lambda$,

$$lK(tu) = \begin{cases} l(2tu) & 0 \le t \le \frac{1}{2}, \\ l(\varphi_{2-2t}(u)), & \frac{1}{2} \le t \le 1 \end{cases}$$

Thus, further splitting cases of t,

$$lK(tu) = \begin{cases} l(2tu) = 4tu & 0 \le t \le \frac{1}{4}, \\ l(2tu) = \varphi_{4t-1}(u) & \frac{1}{4} \le t \le \frac{1}{2}, \\ l(\varphi_{2-2t}(u)) = \varphi_{2-1}(\varphi_{2-2t}(u)) = \varphi_1(\varphi_{2-2t}(u)) \sim \varphi_{2-2t}(u) & \frac{1}{2} \le t \le 1 \end{cases}$$

Define a homotopy $\xi_\tau : X \cup_{\varphi_0} e^\lambda \longrightarrow X \cup_{\varphi_0} e^\lambda$ as follows:

$$\xi_\tau(x) = x, \quad x \in X,$$

$$\xi_\tau(tu) = \begin{cases} (4-3\tau)\, tu, & 0 \le t \le \frac{1}{4-3\tau}, \\ \varphi_{(4-3\tau)t-1}(u), & \frac{1}{4-3\tau} \le t \le \frac{2-\tau}{4-3\tau}, \\ \varphi_{\frac{1}{2}(4-3\tau)(1-t)}(u), & \frac{2-\tau}{4-3\tau} \le t \le 1. \end{cases}$$

At $\tau = 0$, $\frac{1}{4-3\tau} = \frac{1}{4}$, $\quad \frac{2-\tau}{4-3\tau} = \frac{2}{4} = \frac{1}{2}$. Thus

$$\begin{gathered} (4-3\tau)tu = 4tu \\ \varphi_{(4-3\tau)t-1}(u) = \varphi_{4t-1}(u) \\ \varphi_{\frac{1}{2}(4-3\tau)(1-t)}(u) = \varphi_{2(1-t)}(u) \end{gathered}$$

This matches exactly with the definition of lK, thus $\xi_0 = lK$.

At $\tau = 1$, $\quad \frac{1}{4-3\tau} = 1$, hence we are only concerned with the first case, $0 \le t \le 1$,

$$\xi_1(tu) = (4-3\tau) \cdot tu = tu,$$

therefore, $\xi_1 = \text{id}$.

Since ξ_t is a homotopy equivalence, id and lK are homotopic. Moreover, an analogous construction shows that Kl is also homotopic to the identity. Therefore, as K has both left and right inverses, K is a homotopy equivalence. □

Lemma 5.6
Suppose $f : X \to Y$ is a homotopy equivalence, and let $\varphi : S^{n-1} \to X$ be an attaching map. Then f induces a homotopy equivalence

$$F : X \cup_\varphi e^n \longrightarrow Y \cup_{f\circ\varphi} e^n.$$

Proof. Firstly, we define $F|_X = f$ and $F|_{e^\lambda} = \mathrm{id}$. Since f is a homotopy equivalence, we can choose a left inverse $g : Y \to X$ such that $g \circ f \simeq \mathrm{id}_X$. Define

$$G : Y \cup_{f\circ\varphi} e^n \to X \cup_\varphi e^n$$

similarly as F, $G|_Y = g$ and $F|_{e^\lambda} = \mathrm{id}$.

Since $g \circ f \simeq \mathrm{id}_X$, this implies $g \circ f \circ \varphi \simeq \varphi$, therefore, by Lemma 5.5, there exists a homotopy equivalence

$$K : X \cup_{g\circ f\circ\varphi} e^\lambda \longrightarrow X \cup_\varphi e^\lambda.$$

Let h_t denote the homotopy between $g \circ f$ and id. then, $K \circ G \circ F(x) = g \circ f(x)$ for $x \in X$, and for $u \in \partial e^\lambda$,

$$\begin{aligned} K(tu) &= 2tu && \text{for } 0 \le t \le \tfrac{1}{2}, \\ K(tu) &= h_{2-2t}(u) && \text{for } \tfrac{1}{2} \le t \le 1 \end{aligned}$$

Define a homotopy $\eta_\tau : X \cup_\varphi e^\lambda \longrightarrow X \cup_\varphi e^\lambda$ as follows:

$$\eta_\tau(x) = h_\tau(x), \quad x \in X,$$

$$\eta_\tau(tu) = \begin{cases} \frac{2}{1+\tau} tu & 0 \le t \le \frac{1+\tau}{2}, \\ h_{2-2t+\tau}(u), & \frac{1+\tau}{2} \le t \le 1. \end{cases}$$

At $\tau = 0$, this is precisely $\eta_0 = K \circ G \circ F$. At $\tau = 1$, $\frac{1+\tau}{2} = \frac{2}{2} = 1$, thus we are only concerned with the first case with $\eta_1(tu) = tu$ and $\eta_1(x) = h_1(x) = x$, thus $\eta_1 = \mathrm{id}$. Therefore, η_τ is a homotopy from $K \circ G \circ F$ to id, and F has a left homotopy inverse, $K \circ G$.

By a similar construction, we also conclude that G also has a left homotopy inverse.

Assertion 5.7
If $F : X \to Y$ has a left homotopy inverse L, and a right inverse R, then F is a homotopy equivalence and L, R are 2-sided inverses.

$L \circ F \simeq \mathrm{id}$ and $F \circ R \simeq \mathrm{id}$ implies $L \simeq (L \circ F) \circ R = L \circ (F \circ R) \simeq R$, thus $R \circ F \simeq L \circ F \simeq \mathrm{id}$ and R (and L) is 2-sided. This proves Assertion 5.7.

To conclude the proof, $K \circ (G \circ F) \simeq \mathrm{id}$ and K being a homotopy equivalence implies that $(G \circ F) \circ K \simeq \mathrm{id}$. This implies $G \circ (F \circ K) \simeq \mathrm{id}$, which in addition to G having a left inverse and Assertion 5.7, we have $(F \circ K) \circ G \simeq \mathrm{id}$. Again this implies $F \circ (K \circ G) \simeq \mathrm{id}$, and since $(K \circ G)$ is a left homotopy inverse of F, we finally conclude that F is a homotopy equivalence. $\square$

Theorem 5.8 (CW-decomposition of a compact manifold M)
Let $f : M \to \mathbb{R}$ be a Morse function on a compact manifold M. Then M has the homotopy type of a CW-complex having exactly one cell of dimension λ for each critical point of index λ.

Proof. Let $c_1 < c_2 < \cdots < c_r$ be the critical values of f. Since M is compact and critical points are isolated, $\{c_i\}$ can not contain cluster points. Choose $\epsilon > 0$ sufficiently small so that each interval

$$[c_i - \epsilon, c_i + \epsilon]$$

contains only one critical value. For $a < c_1$, M^a is vacous, thus trivially has the homotopy type of a CW complex of dimension -1. For arbitrary c_i, suppose for induction that $M^{c_{i-1}+\epsilon}$ has homotopy type of a CW-complex K, with the homotopy

$$h^{i-1,+} : M^{c_{i-1}+\epsilon} \to K.$$

By the results of Chapter 4, $M^{c_i+\epsilon}$ has homotopy type

$$M^{c_i-\epsilon} \cup_{\phi_1} e^{\lambda_1} \cup_{\phi_2} \ldots \cup_{\phi_{j(c)}} e^{\lambda_{j(c)}},$$

where $j(c)$ is the total number of critical points and each λj denotes the index of each critical point. Since $f^{-1}[c_{i-1} + \epsilon, c_i - \epsilon]$ has no critical points, by Thm 3.3, we have a homotopy $g^i : M^{c_i-\epsilon} \to M^{c_{i-1}+\epsilon}$.

Therefore, each

$$h' \circ h \circ \phi_j : \partial e^{\lambda_j} \xrightarrow{\phi_j} M^{c_i-\epsilon} \xrightarrow{g^i} M^{c_{i-1}+\epsilon} \xrightarrow{h^{i-1,+}} K$$

is homotopic by cellular approximation to

$$\psi_j : \partial e^{\lambda_j} \to (\lambda_j - 1)\text{-skeleton of K}$$

for gluing maps $\phi_j : c^{\lambda_j} \to$ skeleton of K. Therefore, by Lemmas 5.5 and 5.6,

$$K \cup_{\phi_1} e^{\lambda_1} \cup_{\phi_2} \ldots \cup_{\phi_{j(c)}} e^{\lambda_{j(c)}}$$

is a CW-complex with the same homotopy type as $M^{c_i+\epsilon}$.

By induction, beginning with the smallest critical value, we construct M by attaching one cell for each critical point. If M is compact, the only have a finite number of critical values and critical points, therefore this handle attachment process terminates in finite steps with M having homotopy type of a CW-complex with exactly one λ-cell for each critical point of index λ. This concludes the proof of Thm 5.8. □

Remark 5.9 (Non-compact cases)
In the case where M is not compact but still has finitely many critical points, then they all must lie in one of the M^a's. By a similar proof to Thm 3.3, M^a is a deformation retract of M, thus the proof still holds.

We exclude the case where M is not compact and has infinitely many critical points, however, a decomposition still exists in this case. The details can be found in [2, P.23].

Corollary 5.10
In the process of the proof above, we also proved that each M^{a_i} has homotopy type of a finite *CW-complex with 1 cell of dimension λ for each critical point of index λ in M^a, including a if a itself is a critical value.*

Remark 5.11 (Extension to Handlebody Decompositions)
In particular, the above argument admits a geometric interpretation in terms of handlebody decompositions. Instead of associating each critical point of index λ to a λ-cell as in Thm 5.8, we instead associate it with an n-dimensional λ-handle. This is again just a "thickening" of the λ-cell so that the dimension is compatible with the dimension of the manifold in order for the attachments to be performed smoothly.

Therefore, as the parameter a crosses a single critical value of index λ, the sublevel set M^a changes, up to homotopy, by the attachment of a λ-handle, that is, a copy of

$$D^\lambda \times D^{n-\lambda}$$

glued along its attaching sphere,

$$\partial D^\lambda \times D^{n-\lambda}.$$

As a result, M admits a handle decomposition in which exactly one λ-handle is attached for each critical point of index λ.

Moreover, by Thm 2.5, one may choose f to be a self-indexing Morse function*. In this setting, the handle decomposition is particularly transparent. We can reconstruct M by starting with the* 0*-handles ($D^0 \times D^n$), and then successively attach* 1*-handles, then* 2*-handles, and so on, up to n-handles in increasing order of index.*

5.1 Handlebody Decomposition for 3-Manifolds

In dimension $n = 3$, the possible handles are of index $0, 1, 2, 3$. The 0- and 3-handles are just balls D^3, and the 1- and 2-handles are copies of $D^1 \times D^2$ and $D^2 \times D^1$ respectively. Recall the illustrations of 1- and 2- handles in Fig 6. A handle decomposition of a closed orientable 3-manifold M can always be taken to have a single 0-handle and a single 3-handle, so the interesting data lies entirely in how the 1- and 2-handles are attached.

After attaching the 0-handle and all 1-handles, we obtain a handlebody H_1 of some genus g, whose boundary $\partial H_1 = \Sigma_g$ is a closed, orientable, connected surface of genus g, in particular, ∂H_1 is a genus g surface bounding solid genus g handlebody on either side of the surface. This actually gives a decomposition of the surface as a *Heegaard splitting*,

$$M = H_1 \cup_{\Sigma_g} H_2,$$

and we call the common boundary Σ_g the *Heegaard surface*.

5.2 Handlebody Decomposition for 4-Manifolds

In dimension $n = 4$, the handles are of index $0, 1, 2, 3, 4$. As in the 3-dimensional case, the 0- and 4-handles are simply copies of D^4, and a handle decomposition of a closed orientable 4-manifold M can always be arranged to have a single 0-handle and a single 4-handle. The

1-handle $D^1 \times D^3$ and 3-handles $D^3 \times D^1$ are also relatively straightforward. Attaching a 1-handle simply increases the genus of the boundary, and attachment of 3-handles are analogous to the dual/“mirror” of the 1-handle case. Thus the genuinely new and subtle data lies entirely in the 2-handles $D^2 \times D^2$.

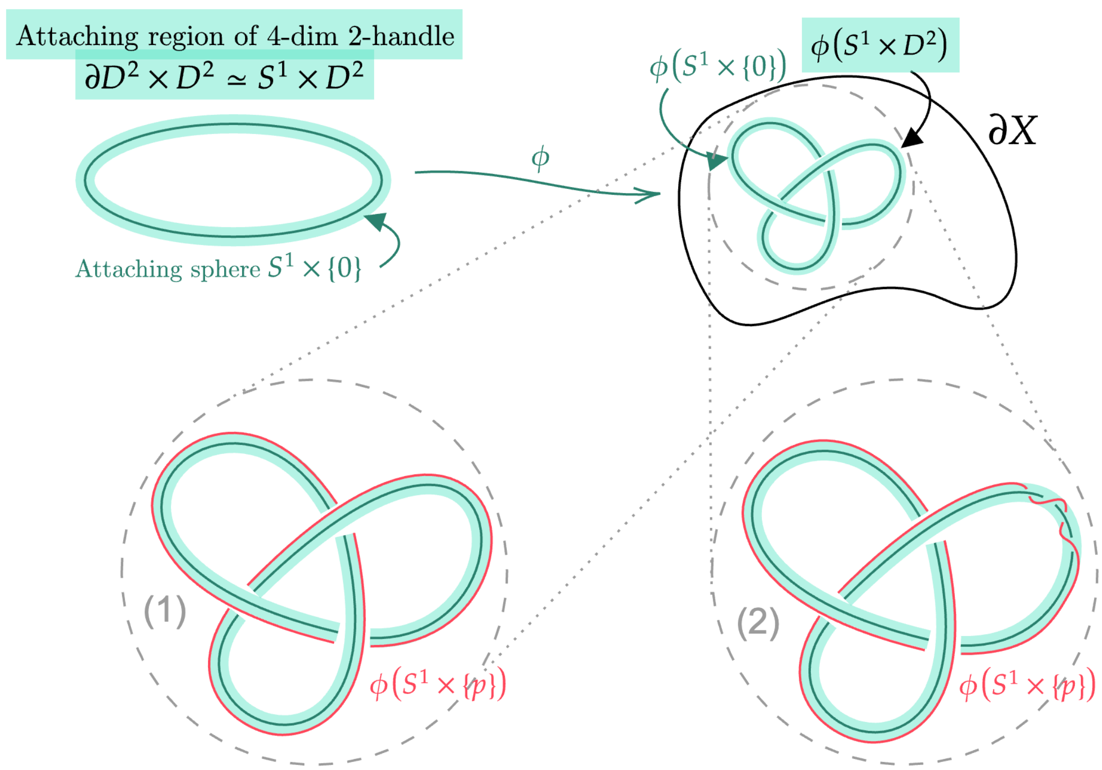


Figure 7: Embedding of a 4-dim 2-handle via the attaching map $\phi : S^1 \times D^2 \to \partial X$, where $dim(\partial X) = 3$. We represent two possible framings of the embedding by choosing a point $p \in D^2$, and visualizing the image $\phi(S^1 \times \{p\})$ in red. Framing (2) contains two “twists” about the image of the attaching sphere, whereas framing (1) has no “twist”. The twisting of $\phi(S^1 \times \{p\})$ represents the internal twisting of $\phi(S^1 \times D^2)$. Therefore, the difference in these two framings is $\pm 2 \in \mathbb{Z} = \pi_2(SO(3))$ (sign depends on the chosen orientation).

2-handles have attaching regions $\partial D^2 \times D^2 \cong S^1 \times D^2$, which is just a solid tori contained in the boundary 3-manifold $\partial(M^a)$. Therefore, the attachment is specified is by two pieces of data. First, the embedding of the attaching circle $S^1 \times \{0\} \hookrightarrow \partial(M^a)$ into an embedded knot or link component, as demonstrated in Fig 7. The second piece of data is the framing of the embedding, which determines the amount and direction of “twists” of the attaching region, relative to the preferred normal direction, this is determined by an integer $n \in \mathbb{Z}$.

More generally, for an n-dim k-handle, the homotopy groups of the special orthogonal group, $\pi_{k-1}(SO(n-k))$, acts on the set of possible framings of the attachment. That is, the difference between any two framings is uniquely determined by an element of $\pi_{k-1}(SO(n-k))$.

For a 3-dim manifold, the homotopy groups associated to 1- and 2-handles are

$$\pi_0(SO(2)) = 0$$
$$\pi_1(SO(1)) = 0$$

respectively, thus both are attached uniquely. Reconsider the case of a 4-dim manifold, the homotopy groups associated to 1-, 2-, and 3- handles are

$$\begin{aligned}\pi_0(SO(3)) &= 0\\ \pi_1(SO(2)) &\cong \mathbb{Z}\\ \pi_2(SO(1)) &= 0\end{aligned}$$

respectively. Thus 1- and 3-handles are attached uniquely, while 2-handles have a non-trivial framing $\mathbb{Z}$ just as mentioned above.

In 5-dim manifolds however, we observe that the homotopy groups associated to 1-, 2-, 3-, and 4-handles are

$$\begin{aligned}\pi_0(SO(4)) &= 0\\ \pi_1(SO(3)) &\cong \mathbb{Z}_2\\ \pi_2(SO(2)) &= 0\\ \pi_3(SO(1)) &= 0\end{aligned}$$

respectively. Conducting a similar analysis on manifolds of higher dimensions illustrates more complex patterns that eventually stabilize, forming the basis for the classification of higher-dimensional manifolds.

6 Morse Inequalities

The Morse inequalities are a collection of inequalities relating the topology of a smooth manifold M to the critical point data of any Morse function $f : M \to \mathbb{R}$ on it. Concretely, they bound a topological invariant, the λ-th Betti number, in terms of analytical/geometric data, the critical points of a index λ. The weak inequalities show that homology can not exceed the critical point count, whereas the strong inequalities refine this with alternating sum bounds.

Its power lies in the two-way transfer between topology and geometry. Knowing the topological structure of a manifold gives lower bounds on the number of critical points that must exist, while information on the Morse function forces upper bound constraints on the homology of M.

Definition 6.1 (Subadditive and Additive Function)
Let S be a function from certain pairs of spaces to $\mathbb{Z}$, S is subadditive if for $Z \subset Y \subset X$, we have

$$S(X, Z) \leq S(X, Y) + S(Y, Z).$$

S is additive if this is an equality.

Corollary 6.2
If S is subadditive, then for $X_0 \subset X_1 \subset \cdots \subset X_n$,

$$S(X_0, X_n) \leq \sum_{i=1}^{n} S(X^i, X^{i-1}).$$

If S is additive,

$$S(X_0, X_n) = \sum_{i=1}^{n} S(X^i, X^{i-1}).$$

If we take $X_0 = \emptyset$, we have

$$S(X) \leq \sum_{i=1}^{n} S(X^i, X^{i-1}), \quad \text{or} \quad S(X) = \sum_{i=1}^{n} S(X^i, X^{i-1}).$$

Betti numbers and the Euler characteristic are examples of subadditive and additive functions respectively.

Definition 6.3 (Betti Number)
The λ-th Betti Number of (X, Y), denoted $R_\lambda(X, Y)$ is the rank of $H_\lambda(X, Y; \mathbb{F})$ over $\mathbb{F}$, for any pair (X, Y) such that this rank is finite.

Definition 6.4 (Euler Characteristic)
The Euler Characteristic of (X, Y) is defined as

$$\chi(X, Y) = \sum (-1)^\lambda R_\lambda(X, Y).$$

Lemma 6.5
The Betti number is subadditive.

Proof. For $Z \subset Y \subset X$, the triple (X, Y, Z) induces SES:

$$0 \to C_*(Y,Z) \xrightarrow{\phi} C_*(X,Z) \xrightarrow{\psi} C_*(X,Y) \to 0$$

where ϕ is induced by inc $V \hookrightarrow X$, and ψ is induced by the quotient map $X/Z \to X/Y$.

This SES induces a LES:

$$\cdots \to H_n(Y,Z) \xrightarrow{i_*} H_n(X,Z) \xrightarrow{j_*} H_n(X,Y) \xrightarrow{\partial} H_{n-1}(Y,Z) \cdots$$

where i_* is induced by the inclusion $(Y,Z) \hookrightarrow (X,Z)$, and j_* is induced by the inclusion $(X,Z) \hookrightarrow (X,Y)$, and $H_i(A,B) = \dfrac{\operatorname{im} \partial_{i+1}}{\ker \partial_i}$ is the relative homology of (A,B).

Then, for

$$\cdots \to A \xrightarrow{\phi} B \xrightarrow{\psi} C \to \cdots$$

an exact sequence,

$$\begin{aligned} \dim B &= \underbrace{\dim(\operatorname{im}\phi)}_{\leq \dim C} + \underbrace{\dim(\ker\psi)}_{=\dim(\operatorname{im}\phi) \leq \dim A} \\ &\leq \dim A + \dim C \end{aligned}$$

Therefore,

$$\underbrace{\dim H_n(X,Z)}_{B_n(X,Z)} \leq \underbrace{\dim H_n(X,Y)}_{B_n(X,Y)} + \underbrace{\dim H_n(Y,Z)}_{B_n(Y,Z)}$$

□

Lemma 6.6
The Euler characteristic is additive.

Theorem 6.7 (The Weak Morse Inequalities)
Let C_λ denote the number of critical points of index λ. Then

$$R_\lambda(M) \leq C_\lambda$$
$$\sum (-1)^\lambda R_\lambda(M) = \sum (-1)^\lambda C_\lambda$$

Proof. We first introduce two theorems that are referenced in this proof.

Theorem 6.8 (Existence of Smooth Bump Functions ([1, Lemma 2.25]))
Let M be a smooth manifold with or without boundary, For any closed subset $A \subseteq M$ and any open subset U containing A, there exists a smooth bump function $\psi : M \to \mathbb{R}$ for A supported in U, such that,

(1) $0 \leq \psi \leq 1$ *on* M
(2) $\psi \equiv 1$ *on* A
(3) $\operatorname{supp}(\psi) \subset U$

Theorem 6.9 (Excision Theorem)
Let (X, A) be a pair of spaces so that $A \subseteq X$. Let U be a subspace of both A and X so that $U \in \operatorname{int}(A)$, the inclusion $(X \setminus U, A \setminus U) \subseteq (X, A)$ is called an excision, and induces an isomorphism in the relative homology,

$$H_*(X \setminus U, A \setminus U) \cong H_*(X, A).$$

Let M be a compact manifold. By Thm 2.5, there exists a self-indexing Morse function, $g : M \to \mathbb{R}$ on M.

Construct a new Morse function from g with the following process. Let P_λ denote the set of critical points of index λ of g, assign each point in P_λ an index $1, 2, \ldots, C_\lambda$. For each $p_i \in P_\lambda$, pick a compact neighborhood $U_{p_i} \subset M$ containing p_i. By Thm 6.8 we can define a function f_i such that

$$f_i(p_i) = \frac{1}{i+1}, \quad f_i'(p_i) = 0, \quad f_i''(p_i) = 0$$

and f_i is compactly supported on U_{p_i}, thus f_i vanishes outside of U_{p_i}. $f_i'(p_i) = 0$ ensures that the point p_i remains a critical point under the summation, while $f_i''(p_i) = 0$ ensures that the Hessian at p_i remains non-zero, thus preserving its non-degeneracy. Therefore, taking the sum of all such f_i's for all λ's with g and by an application of the Partition of Unity Theorem (3.2), we obtain a Morse function $f : M \to \mathbb{R}$ such that critical points of f are the exact same as that of g, with the same indices. However, the critical values of f are distinct, with critical points appearing in order of increasing index.

Let $a_1 < \cdots < a_n$ be the critical values of f. By construction, M^{a_i} contains exactly i critical points, with $M^{a_n} = M$. Consider the relative homology of $(M^{a_i}, M^{a_{i-1}})$, since $M^{a_i} \simeq M^{a_{i-1}} \cup e^{\lambda_i}$,

$$\begin{aligned} H_*\left(M^{a_i}, M^{a_{i-1}}\right) &= H_*\left(M^{a_{i-1}} \cup e^{\lambda_i}, M^{a_{i-1}}\right) \\ &= H_*\left(e^{\lambda_i}, \partial e^{\lambda_i}\right) \\ &= \begin{cases} \mathbb{F}, & \text{for } * = \lambda_i \\ 0, & \text{otherwise,} \end{cases} \end{aligned}$$

where the second equality follows from the Excision Theorem (Thm 6.9), in which we excise with $U = M^{a_{i-1}} \setminus \partial e^{\lambda_i}$.

Therefore, since $R_\lambda(M^{a_i}, M^{a_{i-1}}) = 1$ if and only if $M^{a_i} \simeq M^{a_{i-1}} \cup e^\lambda$, and R_λ is subadditive, we have the first inequality,

$$R_\lambda(M) \leq \sum_{i=1}^{n} R_\lambda(M^{a_i}, M^{a_{i-1}}) = C_\lambda.$$

As for the second relation,

$$\chi(M^{a_i}, M^{a_{i-1}}) = \sum_{\lambda \geq 0} (-1)^\lambda R_\lambda(M^{a_i}, M^{a_{i-1}})$$

is an alternating sum with addition for even dimensions and subtraction for odd dimensions of Betti numbers of the pair, $(M^{a_i}, M^{a_{i-1}})$. Therefore,

$$\begin{aligned}\chi(M) &= \sum_{i=1}^{n} \chi(M^{a_i}, M^{a_{i-1}}) \\ &= \sum_{\lambda \geq 0} (-1)^\lambda R_\lambda(M^{a_i}, M^{a_{i-1}}) \\ &= C_0 - C_1 + C_2 - \cdots \pm C_n.\end{aligned}$$

Thus we have shown

$$\sum (-1)^\lambda R_\lambda(M) = \sum (-1)^\lambda C_\lambda.$$

□

Lemma 6.10
$S_\lambda(X,Y) = R_\lambda(X,Y) - R_{\lambda-1}(X,Y) + R_{\lambda-2}(X,Y) - \cdots \pm R_0(X,Y)$ is subadditive.

Proof. Let $\cdots \xrightarrow{h} A \xrightarrow{i} B \xrightarrow{j} C \xrightarrow{k} \cdots \to D \to 0$ be exact seq of vector spaces. By construction, $\mathrm{rank}(h) + \mathrm{rank}(i) = \mathrm{rank}(A)$. This implies,

$$\begin{aligned}\mathrm{rank}(h) &= \mathrm{rank}(A) - \mathrm{rank}(i) \\ &= \mathrm{rank}(A) - \mathrm{rank}(B) + \mathrm{rank}(j) \\ &\vdots \\ &= \mathrm{rank}(A) - \mathrm{rank}(B) + \mathrm{rank}(C) - \cdots \pm \mathrm{rank}(D)\end{aligned}$$

where the last term is greater than 0 since $\mathrm{rank}(h) \geq 0$.

Consider the homology exact seq of $X \supset Y \supset Z$,

$$\cdots \to H_{\lambda+1}(X,Y) \xrightarrow{\partial} H_\lambda(Y,Z) \to H_\lambda(X,Z) \to H_\lambda(X,Y) \to \cdots$$

applying the above computation to this exact sequence, we have $\mathrm{rank}(\partial) = R_\lambda(Y,Z) - R_\lambda(X,Z) + R_\lambda(Y,Z) - R_{\lambda-1}(Y,Z) + \cdots \geq 0$. Collecting the terms of each $R_*(A,B)$, we obtain,

$$S_\lambda(Y,Z) - S_\lambda(X,Z) + S_\lambda(X,Y) \geq 0.$$

Therefore,

$$S_\lambda(X,Z) \leq S_\lambda(X,Y) + S_\lambda(Y,Z).$$

□

Theorem 6.11 (The Morse Inequalities)

$$R_\lambda(M) - R_{\lambda-1}(M) + \cdots \pm R_0(M) \leq C_\lambda - C_{\lambda-1} + \cdots \pm C_0$$

Proof. Applying Lemma 6.10 to $\emptyset \subset M^{a_1} \subset M^{a_1} \subset \cdots \subset M^{a_k} = M$ as defined in the proof, we obtain

$$\begin{aligned} R_\lambda(M) - R_{\lambda-1}(M) + \cdots \pm R_0(M) &= S_\lambda(M) \\ &\le \sum_{i=1}^{n} S_\lambda(M^{a_i}, M^{a_{i-1}}) \\ &= \sum_{i=1}^{n} \left(\sum_{j=0}^{\lambda} (-1)^j R_j(M^{a_i}, M^{a_{i-1}}) \right) \\ &= \sum_{j=0}^{\lambda} (-1)^j \left(\sum_{i=1}^{n} R_j(M^{a_i}, M^{a_{i-1}}) \right) \\ &= C_\lambda - C_{\lambda-1} + \cdots \pm C_0 \end{aligned}$$

□

The Morse equalities are certainly sharper because they imply the weak Morse inequalities. In particular, if we apply the Morse inequalities for $\lambda, \lambda - 1, \ldots, 0$,

$$\begin{aligned} R_\lambda(M) - R_{\lambda-1}(M) + \cdots \pm R_0(M) &\le C_\lambda - C_{\lambda-1} + \cdots \pm C_0 \\ + R_{\lambda+1}(M) - R_{\lambda+2}(M) \mp R_0(M) &\le + C_{\lambda-1} - C_{\lambda-2} \cdots \mp C_0 \\ &\vdots \\ \pm R_0(M) &\le \pm C_0 \end{aligned}$$

and sum them together, we obtain the weak inequalities

$$R_\lambda(M) \le C_\lambda.$$

As an application of the Morse inequalities, we can prove the following corollary that otherwise can be obtained using Thm 5.8.

Corollary 6.12
If $C_{\lambda+1} = C_{\lambda-1} = 0$, then $R_{\lambda+1} = R_{\lambda-1} = 0$ and $R_\lambda = C_\lambda$

Proof. Suppose $C_{\lambda+1} = 0$, then $R_{\lambda+1} = 0$ since $R_{\lambda+1} \le C_{\lambda+1}$. Applying the Morse inequality for λ and $\lambda + 1$, we obtain

$$R_\lambda(M) - R_{\lambda-1}(M) + \cdots \pm R_0(M) \le C_\lambda - C_{\lambda-1} + \cdots \pm C_0,$$

and

$$\begin{aligned} &\cancelto{0}{R_{\lambda+1}(M)} - R_\lambda(M) + \cdots \mp R_0(M) \le \cancelto{0}{C_{\lambda+1}} - C_\lambda + C_{\lambda-1} + \cdots \mp C_0 \\ \Rightarrow\ &R_\lambda(M) - R_{\lambda-1}(M) + \cdots \pm R_0(M) \ge C_\lambda - C_{\lambda-1} + \cdots \pm C_0, \end{aligned}$$

respectively, together they imply the two sides are equal.

$$R_\lambda(M) - R_{\lambda-1}(M) + \cdots \pm R_0(M) = C_\lambda - C_{\lambda-1} + \cdots \pm C_0$$

Suppose $C_{\lambda-1} = 0$ as well, then $R_{\lambda-1} = 0$, and by a similar argument, $R_{\lambda-2}(M) - R_{\lambda-3}(M) + \cdots \pm R_0(M) = C_{\lambda-2} - C_{\lambda-3} + \cdots \pm C_0$. We can subsequently subtract the second equality from the first to obtain $R_\lambda = C_\lambda$. □

Alternative proof. To prove this theorem using Thm 5.8, observe that $C_{\lambda+1} = C_{\lambda-1} = 0$ implies that our chain complex is of the form,

$$\cdots \to 0 \xrightarrow{\partial_{\lambda+1}} \mathbb{Z}^{C_\lambda} \xrightarrow{\partial_\lambda} 0 \to \cdots .$$

Therefore,

$$R_\lambda = \mathrm{rank}(H_\lambda) = \mathrm{rank}\left(\frac{\ker\partial_\lambda}{\mathrm{im}\partial_{\lambda+1}}\right) = \mathrm{rank}\left(\frac{\mathbb{Z}^{C_\lambda}}{0}\right),$$

therefore, $R_\lambda = C_\lambda$. $\square$

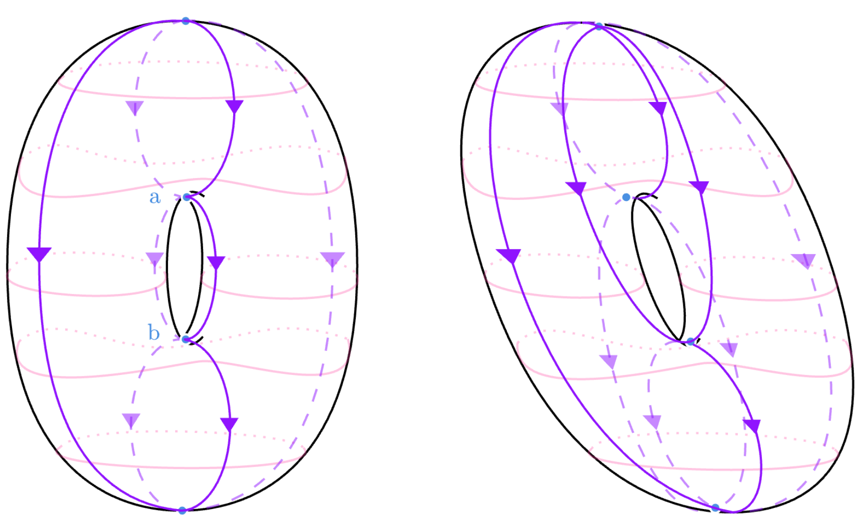
7 Further Explorations: Morse Homology

Morse theory gives us an alternate method of computing homologies of closed manifolds. In particular, we have established the downward gradient of a Morse function f as a vector field, and consider the count of downward gradient flow lines between pairs of critical points x, y of decreasing index as a Moduli space $\hat{M}(x, y)$.

By Thm 2.2, we know that any manifold has a generic set of Morse functions. We further extend this by arguing every pair (M, f) has a generic choice of a Riemannian metric g, which means there exists an open and dense set of metrics which makes (f, g) a Morse-Smale pair. A Morse-Smale pair (f, g) satisfies the condition that ascending (flow lines originate) and descending (flow lines terminate) manifolds of critical points, A^p and D^q intersect transversally. This means, for each $x \in A^p \cap D^q$, the tangent spaces of x in A^p and D^q span its tangent space in M ([1, P.143]),

$$T_x M = T_x A^p + T_x D^q.$$

In particular, we want to avoid situations where ascending and descending intersect tangentially in a way that their tangent spaces coincide or fail to be complementary, thus do not span the full tangent space of the manifold. An explicit instance of this is seen in the following example.



Gradient flow lines of the height function on a vertical torus

Gradient flow lines of the height function on a tilted vertical torus

Figure 8: Height function on the vertical torus and the tilted vertical torus. Blue points are the critical points, pink lines represent contour lines, and finally purple lines represent gradient flow lines between critical points.

As a demonstration, the height function on the torus in Fig 8 with the Euclidean metric inherited from the ambient space is not a Morse-Smale pair. This is due to a lack of transversality. In particular, consider the pair (a, b) of critical points of index 1. A^a consists of the points along the two gradient flow lines between a, b (denote as L) excluding $\{b\}$. Similarly, D^b consists $L \setminus \{a\}$. Therefore, their intersection is $L \setminus \{a, b\}$ and for each point $x \in L \setminus \{a, b\}$, $T_x(L \setminus \{b\})$ and $T_x(L \setminus \{a\})$ coincide, thus this fails to be a transversal intersection.

However, adding a tilt or small perturbation to the torus makes the height function and inherited Euclidean metric a Morse-Smale pair, as it restores transversality.

Furthermore, if (f,g) is a Morse-Smale pair, then for $x \neq y$,

$$\dim \hat{M}(x,y) = \lambda(x) - \lambda(y) - 1.$$

If the index-difference $\lambda(x) - \lambda(y) = 1$, then $\dim \hat{M}(x,y) = 1 - 1 = 0$. Furthermore, we can show $\dim \hat{M}(x,y)$ is compact, and a compact 0-manifold is just a finite set of points, thus we can take the parity of its cardinality $\#\hat{M}(x,y)$.

The finitude of $\#\hat{M}(x,y)$ for pairs of critical points with index difference 1 makes the following map on the Morse-Smale chain complex $CM_*(M,f,g)$ well-defined,

$$\begin{aligned} \partial : CM_*(M,f,g) &\to CM_*(M,f,g) \\ x &\mapsto \sum_{\{y \in \mathrm{Crit}(f) | \lambda(x) - \lambda(y) = 1\}} \#\hat{M}(x,y) \cdot y \end{aligned}$$

To show that this is indeed a chain complex, we must demonstrate that $\partial^2 = 0$. In particular, for pairs (x,y) such that $\lambda(x) - \lambda(y) = 2$, we have $\dim(\hat{M}(x,y)) = 1$ and we can show that the boundary $\partial\hat{M}(x,y)$ consists of broken flow lines between x and y. Note that a broken flow line between x and y is defined as a sequence of trajectories $(\gamma_1, \ldots, \gamma_k)$ that connect x and y through a series of intermediate critical points $(p_1, p_2, \ldots, p_n)$ such that $\lambda(x) - \lambda(p_1) = \lambda(p_i) - \lambda(p_{i+1}) = \lambda(p_n) - \lambda(y) = 1$. In our case, $n = 1$, therefore,

$$\partial^2(x) = \sum_{\{p | \lambda(x) - \lambda(p) = 1\}} \sum_{\{y | \lambda(p) - \lambda(y) = 1\}} \Big(\#\hat{M}(x,p) \cdot \#\hat{M}(p,y)\Big) \cdot y.$$

Since a compact 1-manifold always has an even number of boundary points, in particular S^1 has no boundary and bounded line segments have two boundary points, the sum $\sum \Big(\#\hat{M}(x,p) \cdot \#\hat{M}(p,y)\Big)$ must also be even, i.e. equals 0. Thus $\partial^2 = 0$.

Having defined the chain complex, we can now take the Morse homology

$$HM_k(M,f,g) = \frac{\ker(\partial_k)}{\mathrm{im}(\partial_{k+1})}.$$

Perhaps the most fascinating aspect of Morse homology is the topological invariance and isomorphism to singular homology. Namely, we can show that $HM_*(M,f,g)$ is not only independent of the choice of g, but also independent of the choice of f. Consequently, it is possible to prove that $HM_*(M)$ is isomorphic to singular homology of M,

$$HM_*(M) \cong H_*(M; \mathbb{Z}_2).$$

8 Acknowledgments

I'd like to thank Prof. András Stipsicz for his generous support and mentorship throughout this semester. Our conversations have been a constant source of mathematical inspiration, and I leave this semester having learned a great deal of beautiful mathematics thanks to his guidance. I would also like to thank Balázs Attila Forman and Marco Marengon for their patience with my questions. Finally, I'd like to thank Prof. Peter Ozsváth for his unwavering encouragement from across the Atlantic.

9 Citations

[1] John M. Lee. *Introduction to Smooth Manifolds*. Vol. 218. Graduate Texts in Mathematics. Springer New York, NY, 2012. ISBN: 978-1-4419-9981-8. DOI: https://doi.org/10.1007/978-1-4419-9982-5. URL: https://link.springer.com/book/10.1007/978-1-4419-9982-5.

[2] John Milnor. *Morse Theory*. 1st ed. Annals of Mathematics Studies. Princeton University Press, 1963. ISBN: 978-0691080086. URL: https://webhomes.maths.ed.ac.uk/~v1ranick/papers/milnmors.pdf.

[3] Eddie Nijholt. "The Picard-Lindelof Theorem: Existence and Uniqueness of Solutions". In: (Mar. 2025). URL: https://eddienijholt.nl/Picard-Lindelof.pdf.

[4] Peter S. Ozsváth, András I. Stipsicz, and Zoltán Szabó. *Heegaard Floer homology (Version: December 15, 2025)*. 2025. URL: https://web.math.princeton.edu/~petero/konyv9.pdf.

[5] Andrew Schultz. "Congruence; Sylvester's Law of Inertia". In: (Apr. 2010). URL: https://palmer.wellesley.edu/~aschultz/w10/math416/coursenotes_and_handouts/coursenotes_100422.pdf.